

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- A) 1) Explain the axiomatic definition of probability along with axioms. (L1/CO1) 6
- 2) A card is drawn at random from 52 cards. Find the probability of its being
- i. an Ace
 - ii. a Six or a Heart
 - iii. Neither a Nine nor a Spade

- B) A discrete RV has following probability distribution (L1/CO1) 6

x	0	1	2	3	4	5	6	7	8
P(x)	k	3k	5k	7k	9k	11k	13k	15k	17k

Find the value of k, $P(X < 3)$ and distribution function of X.

- C) A fair coin is tossed 4 times. Define the sample space corresponding to this random experiment. Also give the subsets corresponding to the following events and find the respective probabilities. (L1/CO1) 6
- i. More Heads than Tails are obtained
 - ii. Tails occurred on the even numbered tosses

Q.2 Solve Any Two of the following.

- A) Define the following terms along with examples (L1/CO2) 6
- i. Joint Probability
 - ii. Conditional Probability
 - iii. Independence
- B) A continuous random variable X that can assume any value between $x = 2$ and $x = 5$ has a density function given by $f_X(x) = k(1 + x)$. Find the value of $p(X < 4)$. (L1/CO2) 6
- C) One out of 5 students at a local college say that they skip breakfast in the morning. Find the mean, variance and standard deviation if 10 students are randomly selected. (L2/CO2) 6

Q. 3 Solve Any Two of the following.

- A) Find the covariance matrix for given data $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, covariance matrix = $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. (L1/CO3) 6
- B) In café, customer arrives at rate of 2 per minute. Find the probability of arrival of 5 customers in 1 minute using Poisson distribution. (L2/CO3) 6
- C) Show if given inner product is valid or not. If $\alpha = (a_1, a_2,)$, $\beta = (b_1, b_2,) \in V_2(R)$. (L1/CO3) 6
- Prove: $\langle \alpha, \beta \rangle = a_1 b_1 - a_2 b_1 - a_1 b_2 + 4 a_2 b_2$

Q.4 Solve Any Two of the following.

- A) Prove that if there is convergence in probability then there is convergence in distribution. (L1/CO4) 6
- B) Explain the difference between Weak law of large numbers and Strong law of large numbers. (L1/CO4) 6
- C) Explain in detail with proof Central limit theorem. (L1/CO4) 6

Q. 5 Solve Any Two of the following.

- A) 1) Define Autocorrelation function with all its properties. (L1/CO5) 6
2) If $X(t)$ is WSS process with autocorrelation $R(\tau) = Ae^{-\alpha|\tau|}$ then determine the second order moment of RV $X(8) - X(5)$.
- B) Explain in brief Strict Sense Stationary (SSS) process and Wide Sense Stationary (WSS) process. (L2/CO5) 6
- C) Define random process along with its classification. Also define Mean, Autocorrelation and Autocovariance in terms of random process. (L1/CO5) 6

*** End ***